Structured Query Language:

Structure Query Language(SQL) is a database query language used for storing and managing data in Relational DBMS.SQL was the first commercial language introduced for E.F Codd's Relational model of database. Today almost all RDBMS(MySql, Oracle, Infomix, Sybase, MS Access) use SQL as the standard database query language. SQL is used to perform all types of data operations in RDBMS.

The different categories of SQL used in RDBMS to store and manipulate data are:

- Data Definition Language(DDL)
- Data Manipulation Language(DDL)
- Transaction Control Language(TCL)
- Data Query Language(DQL)

Data Definition Language (DDL)
DDL statements define the data structure such as tables, views etc.. that make up the database. All DDL statements are auto committed means the changes will become permanent and database objects created are available to all users. Commonly used DDL statements are:

- CREATE
- ALTER
- DROP
- RENAME
- TRUNCATE

Data types
A data type identifies or classifies a particular type of information or data. Some commonly used data types are:

- CHAR (size) - Used to store character strings values of fixed length.
- VARCHAR2 (size) – Used to store variable length string data.
- NUMBER (size, precision) – Used to store numbers(fixed or floating point)
- DATE – Used to represent date and time.
- LONG – Used to store large variable length strings(upto 2GB).

DDL statement : CREATE

The CREATE keyword is used for creating database objects like tables, views,triggers, and indexes

Syntax:

CREATE TABLE table_name
(
column_name1 DATATYPE(Size) ,
column_name2 DATATYPE(Size),
column_name3 DATATYPE(Size)
);
Ex: Create Table Employee
(
Emp_id number(4) NOT NULL,
Name varchar2(20) NOT NULL,
Salary number(8),
E_Mail varchar2(30),
Country varchar2(20),
);

The above statement creates a table named Employee with columns Emp_id, ame,Salary,E_Mail and Country.

DDL statement : ALTER
Alter statement is used to modify the structure of database or database objects like tables, views etc..

Alter table statement - to add column to a table:

The above statement adds a new column 'age' of number data type with constraint not null.

Modifying the column
Ex: ALTER TABLE Employee MODIFY salary number(10,2);
Table Altered.

Rename and Drop a Column
Using the alter statement we can rename a column and also the drop any column.

The above statements renames the column name salary to em_sal anddrops column age

DDL statements : TRUNCATE , DROP , RENAME

Truncate Table:
Remove all the rows and resets schema of the table.

TRUNCATE TABLE Employee;

Drop table :
Deletes table entirely from the database.
DROP TABLE Employee;

Renaming a Table:
Changing the name of a table.
RENAME Employee to Emp_table;